

Nicholas J. Nolen

Waco, TX | Nicholas.Nolen1@baylor.edu | linkedin.com/in/nicholasnolen1/ | github.com/nicholasnolen1

Citizenship: United States Citizen | **Veterans Preference:** None (0-Point) | **Target Grade:** GS-1550-12

Position Title: Public Notice, Research Computer Scientist (Direct Hire)

Announcement Number: LaRC-26-DE-13053025-BP | **USAJOBS Control Number:** 883318500

EDUCATION & GS-1550 BASIC QUALIFICATIONS

Baylor University

Waco, TX

Master of Science in Computer Science (Focus: Deep Learning and Network Systems)

Expected Graduation: May 2027

- **Graduate Level Coursework:** Machine Learning, Data Communications, Operating Systems, Advanced Database Design.

Baylor University

Waco, TX

Bachelor of Science in Computer Science, GPA: 3.86 / 4.00 (Magna Cum Laude)

Conferred: May 2026

- **Honors & Awards:** Magna Cum Laude, Dean's List (7 semesters), Full Tuition Faculty Scholarship.
- **GS-1550 Math & CS Requirement (30+ Semester Hours Completed):**
 - * *Mathematics & Statistics (15+ Credits):* Calculus I (Differential Calculus), Calculus II (Integral Calculus), Linear Algebra, Probability & Statistics, Discrete Mathematics.
 - * *Computer Science Core (30+ Credits):* Data Structures & Algorithmic Analysis, Systems Programming, Operating Systems, Foundations of Computing, Database Design, Cybersecurity, Software Capstone, Technical Writing.

PEER-REVIEWED PUBLICATIONS

UI-LIC: A Unified Framework for Evaluating Learned Image Compression Models

2026

- **Authors:** Nicholas J. Nolen, Luc Trudeau, and Andrew C. Freeman.
- **Venue & Status:** Accepted for publication in *ACM Multimedia (ACM MM) 2026* (in final picture-ready review).
- **Preprint & Code:** arXiv:2606.23545 [cs.MM] | Open-source code: github.com/BaylorMultimediaLab/UI-LIC (MIT License).
- Built an open-source evaluation framework in Python and PyTorch unifying six learned image compression architectures with HEVC and VVC intra-frame codecs.
- Developed an automated controller that generates Conda environments from model dependency files and manages batch inference jobs through shared JSON configuration schemas.
- Implemented an interactive desktop GUI in PyQt with residual error heatmap overlays to calculate PSNR, SSIM, VMAF, and LPIPS quality metrics.

RESEARCH EXPERIENCE (GS-11 EQUIVALENT)

Baylor University Multimedia Research Laboratory

Waco, TX

Graduate Research Assistant / Lead Developer

January 2026 to Present

Hours: 20 hrs/week (Graduate Research Appointment)

Supervisors: Dr. L. Trudeau / Dr. A. Freeman (May contact)

- Designed the core software architecture for UI-LIC, an open-source evaluation platform resolving environment conflicts across disparate neural image compression models.
- Wrote an experiment controller in Python that dynamically configures Conda environments, reads task configurations, and runs multi-model inference pipelines.
- Built automated evaluation pipelines across benchmark image datasets (Kodak, CLIC, Tecnick), standardizing rate-distortion curves and metric logs.
- Developed an interactive visualization tool with heatmap overlays to inspect localized reconstruction errors and compression artifacts.
- Authored the primary research paper and technical documentation, leading to first-author acceptance at ACM Multimedia 2026.

WORK EXPERIENCE (GS-11 EQUIVALENT)

Vanta ESports
AI Product Engineering Intern
Hours: 40 hrs/week (Full-Time Internship)

Remote
May 2025 to August 2025
Supervisor: Product Engineering Leadership (May contact)

- Built a low-latency telemetry tracking pipeline in **Rust** to process high-frequency in-game coordinate data with sub-millisecond execution times.
- Trained a custom **YOLOv8** object detection model on gameplay video to extract real-time entity coordinates for tactical analysis.
- Connected the Anthropic Claude API to the telemetry stream to generate post-match coaching feedback and gameplay insights.
- Deployed backend services on **Azure** and compiled core Rust tracking modules to **WebAssembly (WASM)** for browser execution via **Vercel**.
- Presented system architecture and benchmarks to the CEO and Board of Directors, leading to the project being featured for a \$1.5M funding proposal.

TECHNICAL PROJECTS

CNN Sputum Smear Tuberculosis Diagnostics Pipeline | *Python, Keras, TensorFlow*

January 2026 to May 2026

- Trained a 5-layer Convolutional Neural Network with batch normalization and data augmentation to classify tuberculosis bacilli from Ziehl-Neelsen stained sputum smears.
- Reached a **0.9198 validation accuracy** and **0.2097 binary cross-entropy loss** across clinical microscopy slide datasets (ODS3 and TBDB).
- Built an automated preprocessing script extracting normalized 224 × 224 pixel crops, and developed an interactive Crop Likelihood Viewer to help technicians spot high-probability fields faster.

Bear Radio: Media Over QUIC (MoQ) Streaming Service | *Rust, MoQ, Ratatui, GCP, Opus*

January 2025 to May 2025

- Built a terminal-based audio streaming service in **Rust** using the Media over QUIC (MoQ) protocol for synchronized playback across distributed devices.
- Designed an asynchronous pub/sub pipeline with **Opus** audio compression and a **Ratatui** terminal UI, deploying instances on **GCP** to test packet pacing and playback under simulated jitter.

MarketOS (Farmers Market Capstone) | *Next.js, Spring Boot, PostgreSQL*

January 2026 to May 2026

- Built a full-stack vendor management web app for local market managers, replacing paper logs with relational database storage.
- Wrote custom financial calculation logic for vendor revenue splits and dynamic spreadsheet generation for grant auditing.

TECHNICAL SKILLS

Languages: Python, Rust, C, C++, Java, SQL (PostgreSQL, MariaDB), Bash, WebAssembly (WASM)
Machine Learning & Data: PyTorch, TensorFlow, Keras, YOLOv8, OpenCV, NumPy, SciPy, PyAV, Scikit-learn
Systems, Cloud & Tools: Linux, Azure, Google Cloud Platform (GCP), Docker, Conda, Git, REST APIs, LaTeX
Competencies: Neural image compression, computer vision, rate-distortion evaluation, network transport protocols

LEADERSHIP & CAMPUS INVOLVEMENT

Baylor University Chamber of Commerce
Secretary, Webmaster, and Bear Committee Member

Waco, TX
August 2023 to December 2025

- Maintained organization web pages using Drupal and handled operational budgets for student events.
- Served on the Bear Committee, feeding and caring for the university’s live American black bear mascots in an on-campus habitat meeting USDA animal welfare standards.